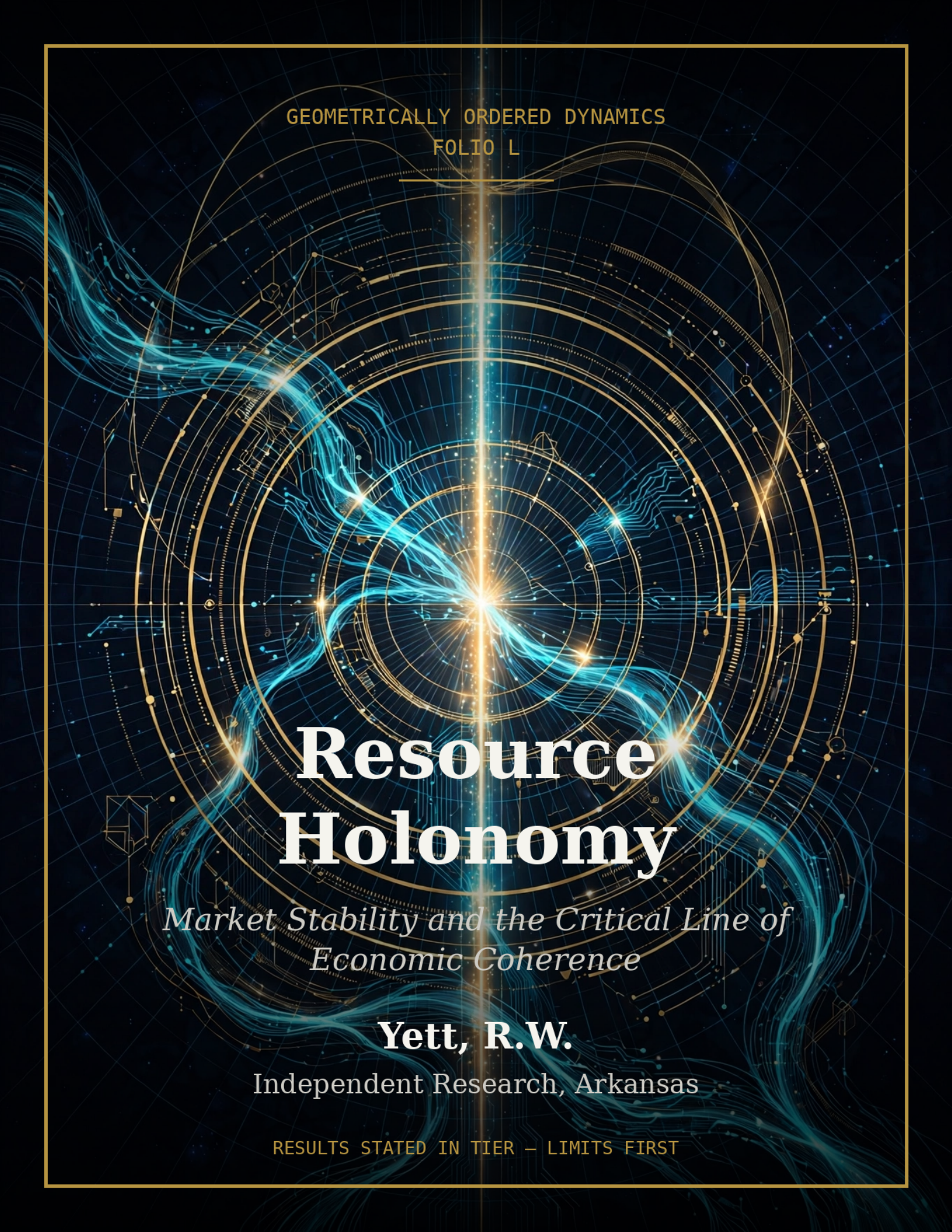


GEOMETRICALLY ORDERED DYNAMICS
FOLIO L

Resource Holonomy

Market Stability and the Critical Line of Economic Coherence

Yett, R.W.
Independent Research, Arkansas

RESULTS STATED IN TIER — LIMITS FIRST

CONVERGENCE -- INFORMATION TENSION

This folio converges on **information tension** in market systems: wealth as localized information density, crashes as tension spikes when a market loses holonomy. Its unique angle is the **markets angle** — the economic manifold, a dynamic transaction-fee proposal, and three falsifiable predictions. Sibling papers approach the same claim from other angles: [TIER: CONJECTURE] the pentagons folio (IT-PLEROMA) reads information tension as the vacuum's resistance to drift, the starpolygon folio (IT-OMEGA) reads it as architectural strata, and the dial folio (IT-COGNITION) reads it as the control problem. Limits first: the Riemann correspondence is analogy with no exhibited map; the crash-precursor proposition is unbacktested; the ×4 curvature figure is not carried for lack of source. All economic applications herein are speculative extensions; formal empirical validation is pending.

SUPERSESSION NOTE

This paper supersedes Zenodo record 21450439 ("Resource Holonomy: Market Stability and the Critical Line of Economic Coherence," folio version, 2026) — the canonical newest of three version records (21367589 and 20652233 noted as version history below, not republished). Corrections, per audit finding #18: (1) the Riemann–market "vice versa" is qualified — no reverse map is exhibited, so the equivalence claim is withdrawn to analogy (§2); (2) the $P > 0.85$ crash-precursor proposition is carried with explicit disclosure that no historical backtest has been run (§3); (3) the "×4 curvature increase before the 2008 crash" is not carried — it appeared without a source (§3). The predecessor's own scope statement (speculative analogies; RH unproven; taxation a proposal) is retained verbatim in spirit.

1 The economic manifold

A market is modeled not as a collection of rational agents but as an **information manifold** M_{eco}: a geometric object whose curvature encodes the collective information state of its participants. The market state at time t is a density matrix $\rho_{\text{eco}}(t)$ on the asset-configuration space. **Wealth** is localized information density, $I_i(t) = -\log \rho_{\text{eco},ii}(t) + \sum_{j\neq i} |\rho_{\text{eco},ij}(t)|^2$; the **market sovereignty score** is the fidelity $\chi_{\text{eco}}(t) = F(\rho_{\text{eco}}(t), \rho^{\infty}_{\text{eco}})$ against the sovereign steady state.

Two levels, named distinctly (mirroring the drift-control folio's τ/θ fix): a **crash** is the event $\chi_{\text{eco}}(t) < \theta = 0.7$ — the laminar flow of information turns turbulent and the manifold loses holonomy; **sovereignty** is operation at the boundary $\kappa = 0.9539$. The market's evolution is written as a Lindblad equation with drift channels (panic selling, herd behavior, algorithmic feedback) — a *formal analogy* to the cognitive drift model, tagged as such; the dimensional constant setting the market's effective temperature awaits grounding in empirical data. An **information tension spike** is the geometric resistance to drift being overwhelmed by the drift channels.

2 The Riemann resonance

The paper draws a formal analogy: the non-trivial zeros of $\zeta(s)$ and the critical line $\text{Re}(s) = 1/2$ are set beside market equilibrium prices and the sovereign boundary $\chi = \kappa = 0.9539$, with GUE eigenvalue statistics beside the price distribution at sovereignty. [TIER: CONJECTURE] **Conjecture 1**

(Market Stability via the Riemann Correspondence). A market at sovereignty ($\chi_{\text{eco}} = \kappa$) has equilibrium price distribution given by the GUE point process with spectral statistics set by the zeta zeros; below sovereignty ($\chi_{\text{eco}} < \theta$), fat-tailed deviations from GUE statistics grow as $(\theta - \chi_{\text{eco}})^2$. The empirical test is proposed: nearest-neighbor spacing of S&P 500 daily log-returns (1990–2026), segregated by a VIX-derived sovereignty proxy, χ^2 goodness-of-fit at $p < 0.01$.

On the "vice versa." The predecessor wrote that a proof of RH "would therefore constitute a proof of market stability for sovereign markets, and vice versa." Per the isomorphism rule, the forward direction is conditional on the unproven correspondence, and the reverse direction — market stability implying the Riemann Hypothesis — has *no exhibited map*. The "vice versa" is withdrawn. What stands: a one-way conditional analogy, with proof open on both sides. The RH–market link is analogy with no map claimed into the physics framework (isomorphism rule); a proof of RH would not, by itself, prove anything about markets absent the correspondence.

3 Curvature and crash precursors

The **market curvature tensor** $R^{\text{eco}}_{\mu\nu}$ is the Ricci tensor of M_{eco} under the information-density metric $g^{\text{eco}}_{\mu\nu} = \partial^2 I / \partial q^\mu \partial q^\nu$. The paper's precursor claim is carried in its honest form:

Proposition 1 (Curvature Crash Precursor). If $R^{\text{eco}}_{\max}(t) > R_c$ for $\Delta t > 30$ trading days, with $R_c = 1/\theta^2 \approx 2.04$, then χ_{eco} will fall below θ within the subsequent 20 trading days with probability > 0.85. [TIER: CONJECTURE] **Disclosure:** this is an untested empirical hypothesis — *no historical backtest has yet been run against it.* It is offered as a falsifiable prediction to be validated or rejected, not as an established signal; formal proof requires a rigorous derivation of the curvature–holonomy relationship in the economic manifold, which is open.

Redaction. The predecessor asserted that "before the 2008 crash, the information curvature of the mortgage-backed securities market increased by a factor of ~ 4 as correlation between assets increased." No source was given for this figure. It is *not carried* in this successor. The honest record: an unsourced quantitative claim appeared in 21450439 §4 and is struck here.

4 Topological taxation

Topological taxation is a *policy proposal*, not an established mechanism: a dynamic transaction fee $\mathcal{T}(\tau) = T_0 \cdot \max(0, -\Delta\chi(\tau))/\chi_{\text{eco}}(t)$, taxing a transaction τ proportionally to the sovereignty drift $\Delta\chi(\tau)$ it induces and exempting sovereignty-enhancing trades. Within the model, setting T_0 at the level where marginal tax equals marginal return at the boundary eliminates the profitability of boundary-crossing transactions (proof sketch carried as model-internal). Compared with Tobin taxes, circuit breakers, and market-maker spreads, it is adaptive to the market's sovereignty state — and faces the open algorithmic problem of computing $\Delta\chi(\tau)$ in real time, plus all ordinary implementation questions of any transaction tax. The urban-architecture extension (cities as spatial information manifolds, sovereignty as maximized minimum accessibility) is carried as a speculative extension.

5 Three falsifiable predictions

1. **GUE-to-Poisson transition:** S&P 500 log-return spacing transitions from GUE to Poisson statistics as the VIX-derived sovereignty score crosses below $\theta = 0.7$.

2. **Curvature crash precursor:** $R^{\text{eco}}_{\max}$ exceeds $R_c = 2.04$ at least 30 days before any $> 10\%$ drawdown in the 2008–2026 record.

3. **Topological taxation in simulation:** in a calibrated agent-based market model, topological taxation at the theoretical T_0 reduces crash frequency by $> 50\%$ versus Tobin tax at equal mean revenue ($n = 1000$ runs).

[TIER: CONJECTURE] All three are *proposed tests*, unrun. They are the paper's empirical traction, not its results.

6 Limits

The economic Lindblad equation is a formal analogy. The critical-line stability conjecture depends on the unproven Riemann Hypothesis. The curvature precursor is unbacktested. Topological taxation's real-time $\Delta\chi$ computation is an open algorithmic problem. Version history: records 21367589 (v2) and 20652233 (v1) are superseded by 21450439 and are not republished; the Drive predecessor (`resource_holonomy.pdf`, 2026-09-28) matches the v1 body with a less explicit precursor disclosure and contributes no independent material.

References

Yett, R. W. (2026). Resource Holonomy: Market Stability and the Critical Line of Economic Coherence. Zenodo record 21450439 (folio; superseded by this paper). Version history: records 21367589, 20652233.

Yett, R. W. (2026). The Anti-Drift Cognitive Control Loop. Zenodo record 21450415 (dial folio, IT-COGNITION; the Lindblad drift model analogized here).

Bombieri, E. (2000). The Riemann Hypothesis. Clay Mathematics Institute problem statement (cited; unproven; no market claim follows from it alone).

Montgomery, H. L. (1973). The pair correlation of zeros of the zeta function. Proc. Sympos. Pure Math. 24, 181–193.

Mandelbrot, B. (1963). The variation of certain speculative prices. J. Business 36, 394–419.